

GREG DICRISTOFARO

Software Engineer

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SUMMARY

Software Engineer with 8+ years of experience in data pipeline processing and former K-12 music **educator**. Proven ability to design, test, and deploy high-performance, resilient applications leveraging cloud services (GCP, AWS) and modern DevOps practices (Docker, Jenkins).

EXPERIENCE

SKILLS

Languages: Python, Bash, SQL, C, C++, TypeScript, Java, C#, Rust, HTML/CSS

Web/Backend: Flask, React, Angular, Spring Boot, ASP.NET Core, Server Sent Events (SSE), WebAssembly

Cloud/DevOps: GCP (BigQuery, Cloud Functions, Datastore), AWS (Lambda, DynamoDB, S3), Azure Blob Storage, Docker, Jenkins, Linux

Databases: PostgreSQL, MS SQL Server, SQLite

Desktop: Tauri, JavaFX, Swing

Development/VCS Tools: Git, GitHub, Claude Code, Copilot, Antigravity, Visual Studio Code, Visual Studio IDE, IntelliJ IDEA IDE

ACHIEVEMENTS

✓ Presenter

Presenter for OSDfCon 2021 (Autopsy Scoring: Finding the Relevant Data with Analysis Results)

✓ KBR Certificate of Appreciation

In recognition of outstanding performance & commitment to excellence (2019)

PROJECTS

constantqjs

Implemented the Constant-Q transform for spectral analysis using C++ compiled to WebAssembly (via Emscripten), integrating with TypeScript/Angular for real-time visualization (Chart.js, Web Audio API)
<https://github.com/gdicristofaro/constantqjs>

Software Engineer

Sleuth Kit Labs

📅 February 2020 - April 2026 © Somerville, MA

- Developed **Digital Forensics and Incident Response software**: Cyber Triage, Autopsy, and The Sleuth Kit in an Agile/Scrum environment
- Expanded and optimized **data pipeline processing** handling complex SQL queries over large data sets using Python, Java, C++
- Created **web crawling and scraping** scripts aggregating open source data for enriching the data pipeline aggregating new features
- Designed and implemented cloud updatable heuristics served to on-premises deployments via **GCP Datastore, BigQuery and Cloud Functions**, then compiled into optimized predicate matching data structures
- **Modernized testing and CI/CD infrastructure** using Jenkins, Docker, Linux, Bash, GitHub Actions, Python, and JUnit/Mockito
- **Standardized multi-platform deployment (Windows, Linux, macOS)** through Snap, Homebrew, and Advanced Installer packages
- Provided internal and **customer documentation** for complex features
- Architected and built **RESTful API and Reactive web applications** using Spring Boot, Swagger, Angular, and React (Material UI), leveraging Server Sent Events (SSE) for real-time data streaming
- Developed parallelized data ingest utilizing **Docker, JMS, ActiveMQ** for coordination
- Utilized AWS DynamoDB, AWS Lambda creating a file hash lookup web service for third-party integration
- Worked with AWS S3 and Azure Blob Storage as data sources for upload and download
- Developed **serverless and containerized cloud infrastructure** using Docker, JMS, ActiveMQ, AWS DynamoDB, AWS Lambda, AWS S3, GCP Cloud Functions, GCP BigQuery, and GCP Datastore
- Developed **Java Desktop Applications** with Swing, JavaFX, Maven, Apache Ivy, Apache POI, SQLite, PostgreSQL, and Netbeans RCP; **Contributed to ControlsFX**, a popular JavaFX controls library

sci-fi-crawl

Sci-fi customizable opening credit text using React, TypeScript, Material UI, and CSS animations <https://github.com/gdicristofaro/sci-fi-crawl>

PSDtoSVG

Python web application for converting Photoshop files into SVG's for data visualization <https://github.com/gdicristofaro/PSDtoSVG>

SeamCarve

Scala implementation of the Seam-Carving Algorithm for content-aware image resizing <https://github.com/gdicristofaro/SeamCarve>

CSPSolver

Class scheduling program based on the AC-3 algorithm utilizing Apache POI API to import from Excel documents written in Scala <https://github.com/gdicristofaro/CSPSolver>

CalendarMaker

Calendar making application developed with TypeScript, React, Moment.js, Material-UI, and PptxGenJS <https://github.com/gdicristofaro/CalendarMaker>

SpellChecker

Spell-checking algorithm built in C checking words against a dictionary file utilizing a self-developed hash map with buckets <https://github.com/gdicristofaro/SpellChecker>

Software Engineer

KBR

📅 July 2017 - February 2020 📍 Lexington Park, MD

- Developed Single Sign On (SSO) and Role-Based Access Control (RBAC) web application
- Queried large datasets using Microsoft SQL Server, T-SQL, Entity Framework, and Linq to SQL
- Created tooling to **generate parameterized queries** improving performance and security
- Created **data visualizations** with JavaScript, TypeScript, **D3.js**, and SVG to present complex data to customers
- Developed and maintained **Web-based UI and RESTful API services** with **React and ASP.NET Core**
- Modernized large J2EE web applications built with JSP and JSF winning an award of excellence for customer satisfaction

Music Teacher

St. Mary's County Public Schools

📅 August 2006 - June 2017 📍 Leonardtown, MD

- Built software for classroom use and tooling to automate school scheduling
- Created school-wide schedules and led public-facing events

EDUCATION

Master of Science in Software Development

Boston University

📅 September 2017 - May 2019 📍 Boston, MA 🎓 GPA 3.96 / 4.00

Software Engineering Certificate

Harvard University

📅 September 2012 - December 2014 📍 Cambridge, MA 🎓 GPA 4.00 / 4.00

Master of Music in Music Education

University of the Arts

📅 June 2012 - December 2015 📍 Philadelphia, PA 🎓 GPA 4.00 / 4.00

- *Designed cross-platform music composition software to foster creativity for young students that processes PCM audio data, visualizes music in a sequencer and renders standard music notation*
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Bachelor of Science in Music Education

Pennsylvania State University

📅 August 2002 - May 2006 📍 University Park, PA 🎓 GPA 3.71 / 4.00